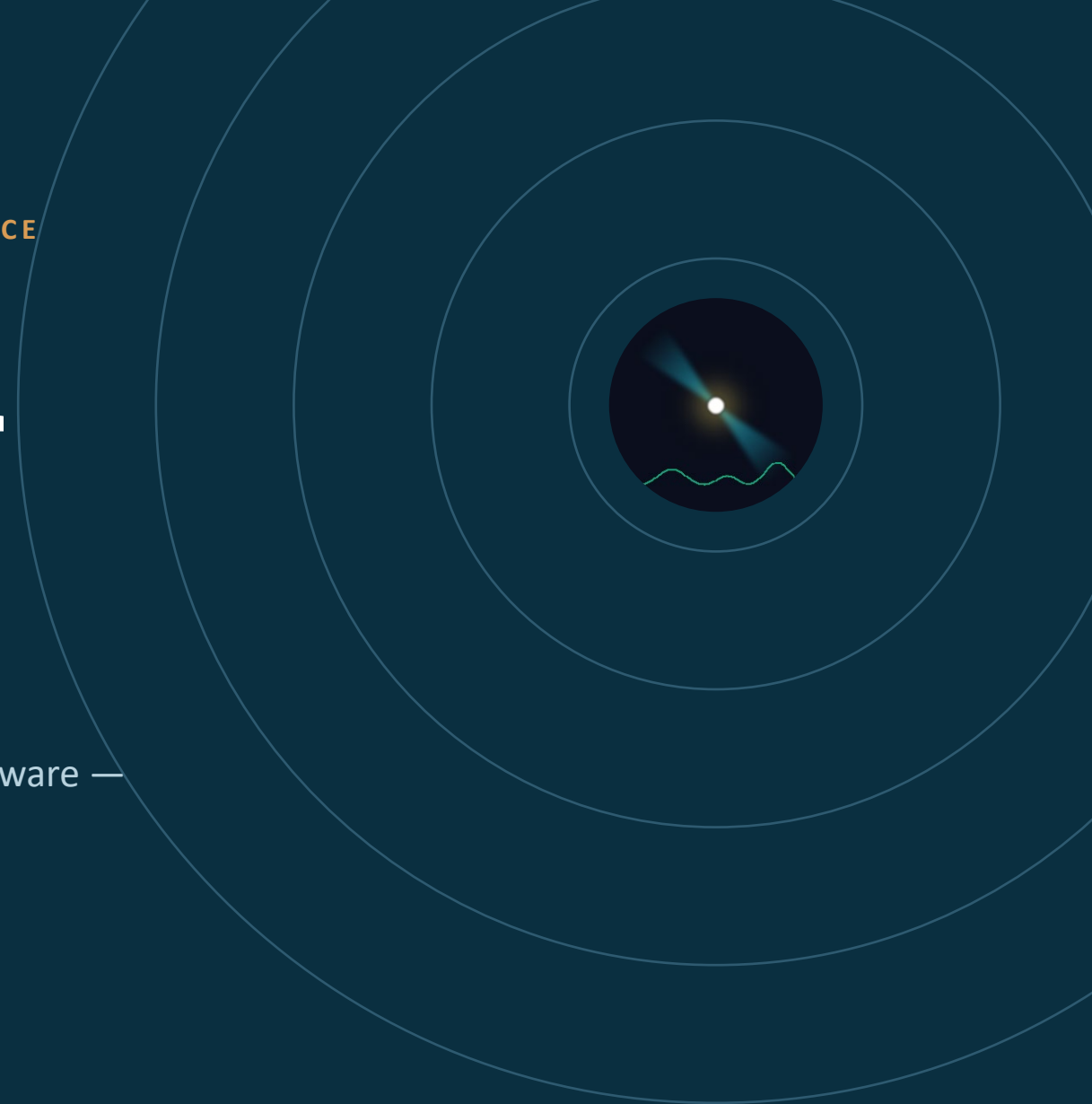


SOCIETY OF AMATEUR RADIO ASTRONOMERS • 2026 CONFERENCE

Lowering the Barrier to Radio Astronomy

First pulsar detections with the new DSES Spectrum Analyzer software —
60-foot dish, Haswell, Colorado

Richard M. Hambly (K0GD) • Deep Space Exploration Society



Abstract

For years, pulsar observing at the Deep Space Exploration Society has rested on a professional toolchain — PRESTO and TEMPO built from source, custom GNU Radio flowgraphs, SIGPROC utilities, and companion analysis tools — with a learning curve steep enough that only two members could take an observation from dish to detection. This talk describes a deliberate experiment in lowering that barrier.

DSES developed the “Spectrum Analyzer”, a cross-platform application that consolidates the on-site workflow — planning the observation, choosing clean frequencies, verifying the RF path, and recording analysis-ready SIGPROC filterbank data — into a single, easy-to-use package driving an Ettus USRP B210 software-defined radio.

On July 11, 2026, a three-member team, none of them pulsar-processing experts, put the premise to the test at the Society’s 60-foot dish in Haswell, Colorado. B0329+54 was ultimately detected at 28 sigma — a figure that grew from 20 when peer review of the trip report exposed, and repaired, a subtle recording-timebase defect — and B0950+08 produced a strong ~9-sigma candidate whose best-fit period and dispersion land on the catalog values.

Field feedback drove two software releases within a week, and a documented upgrade queue — from a built-in pulsar visibility planner to one-click post-recording analysis — now charts the path to training a new generation of DSES observers on a far gentler learning curve.

SPEAKER

Richard M. Hambly, K0GD

SOCIETY

Deep Space Exploration Society (DSES),
Colorado

INSTRUMENT

60-ft dish + Ettus USRP B210 SDR

SOFTWARE

DSES Spectrum Analyzer + PRESTO

KEYWORDS

pulsars • SDR • filterbank • accessibility

Pulsar observing had become an experts-only craft

For years, the Society's results rested on two members — **Dr. Richard Russel (AC0UB)** and **Dan Layne (AD0CY)** — and the professional toolchain they mastered:

PRESTO v5 + TEMPO1 + TEMPO2

built from source; hand-edited
observatory and clock files

RIPTIDE

a second, independent pulsar-search code
(fast-folding)

GNU Radio

custom filterbank flowgraphs driving the
B210

SIGPROC

filterbank-format utilities

blimpy / watutil

Breakthrough Listen RFI and saturation
checks

PINT

high-precision pulsar timing in Python

ATNF pulsar catalogue

ephemeris data (.par) for every fold

Murmur + Best Profile Analyzer

IONAA planning/analysis tools — run under
Wine

Stellarium + experience

above all, knowing how the pieces fit
together

100s

of pages of DSES training
material for this stack

2

members who could run an
observation end to end

PSRCHIVE — the professional community's archiving & timing suite, in common use — beyond even our experts' stack

*First-rate results — but a learning
curve that kept everyone else
watching.*

The new stack — much simpler

The same observation today takes two pieces:

ACQUISITION — ON SITE

DSES Spectrum Analyzer

plan, survey, verify, record — one application on one laptop (Windows / macOS / Linux)



ANALYSIS — ANY MACHINE

PRESTO v6.0.0 + TEMPO2

the professional analysis suite, installed by one DSES build script — v6 barycenters in-process, so classic TEMPO is no longer needed

2

pieces in the whole chain — acquisition and analysis

1

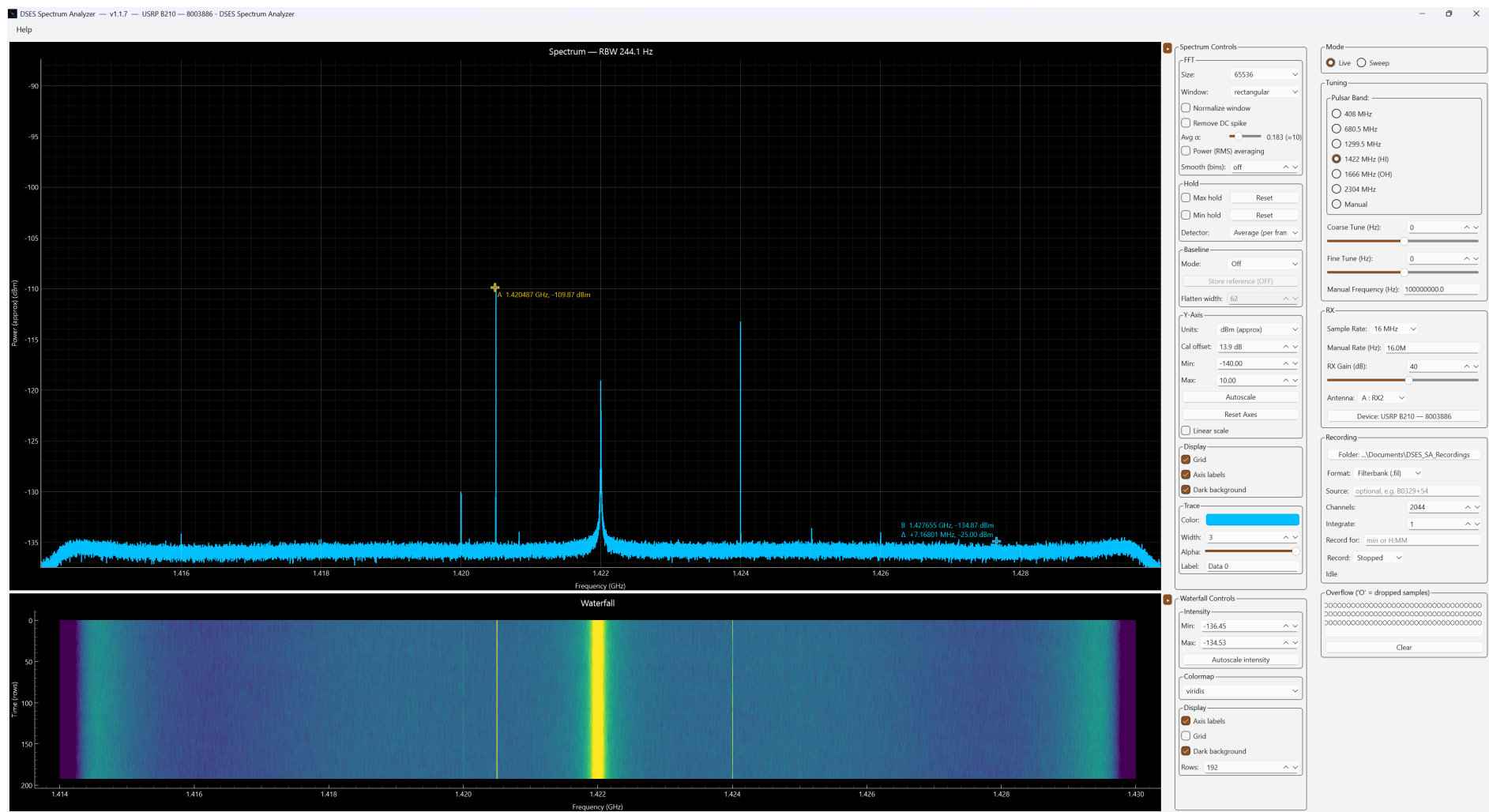
build script installs the analysis stack, per platform

That's the whole list — no Wine, no custom flowgraphs, no hand-edited configuration files

Same professional-quality results — on a far gentler learning curve.

DSES Spectrum Analyzer V1.1.7

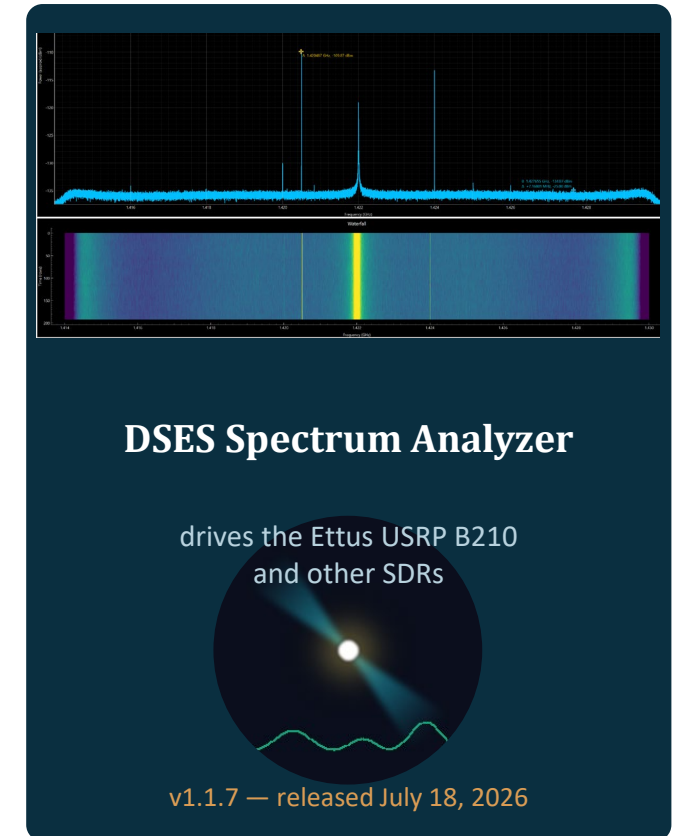
Live B210 at 1420 MHz: three weak test signals above a sub-dB noise floor



One instrument instead of a toolchain

The DSES Spectrum Analyzer — a cross-platform application written within the Society. Born as an RFI-survey instrument for the Haswell site, it has grown into a radio-astronomy data-acquisition system.

- 1 **Plan** — live altitude/azimuth for catalog pulsars, computed from the site
- 2 **Survey** — real-time spectrum and waterfall; wide-band Sweep mode for RFI surveys
- 3 **Verify** — confirm the RF path is alive and clean before committing to a recording
- 4 **Record** — industry-standard SIGPROC .fil files — source name and RA/Dec stamped in the header
- 5 **Update** — built-in updater keeps observatory and member machines on the released baseline



Windows • macOS • Linux
(incl. Raspberry Pi 5 field box)

Put non-experts in front of the dish

***The question:** can members who are not pulsar-processing experts plan an observation, capture pulsar data, and process it into professional-quality reports?*



Ray Uberecken

AA0L



Anne Haney

WOZDW



Richard Hambly

K0GD

60-ft dish

DSES site, Haswell, Colorado

Ettus USRP B210

software-defined radio receiver

Spectrum Analyzer

one laptop — planning, RF checks, recording

Two pulsars recorded live to filterbank

20 MHz

bandwidth, centered on 420 MHz

256

frequency channels

204.8 μ s

time resolution

32-bit

samples — SIGPROC .fil

Target	Start (MDT)	Duration	Data volume	Why this target
B0329+54	14:17	36.6 min	11.0 GB	Brightest northern pulsar — the standard reference source
B0950+08	15:12	22.2 min	6.7 GB	Bright, low-dispersion; transiting at ~59° altitude during the session

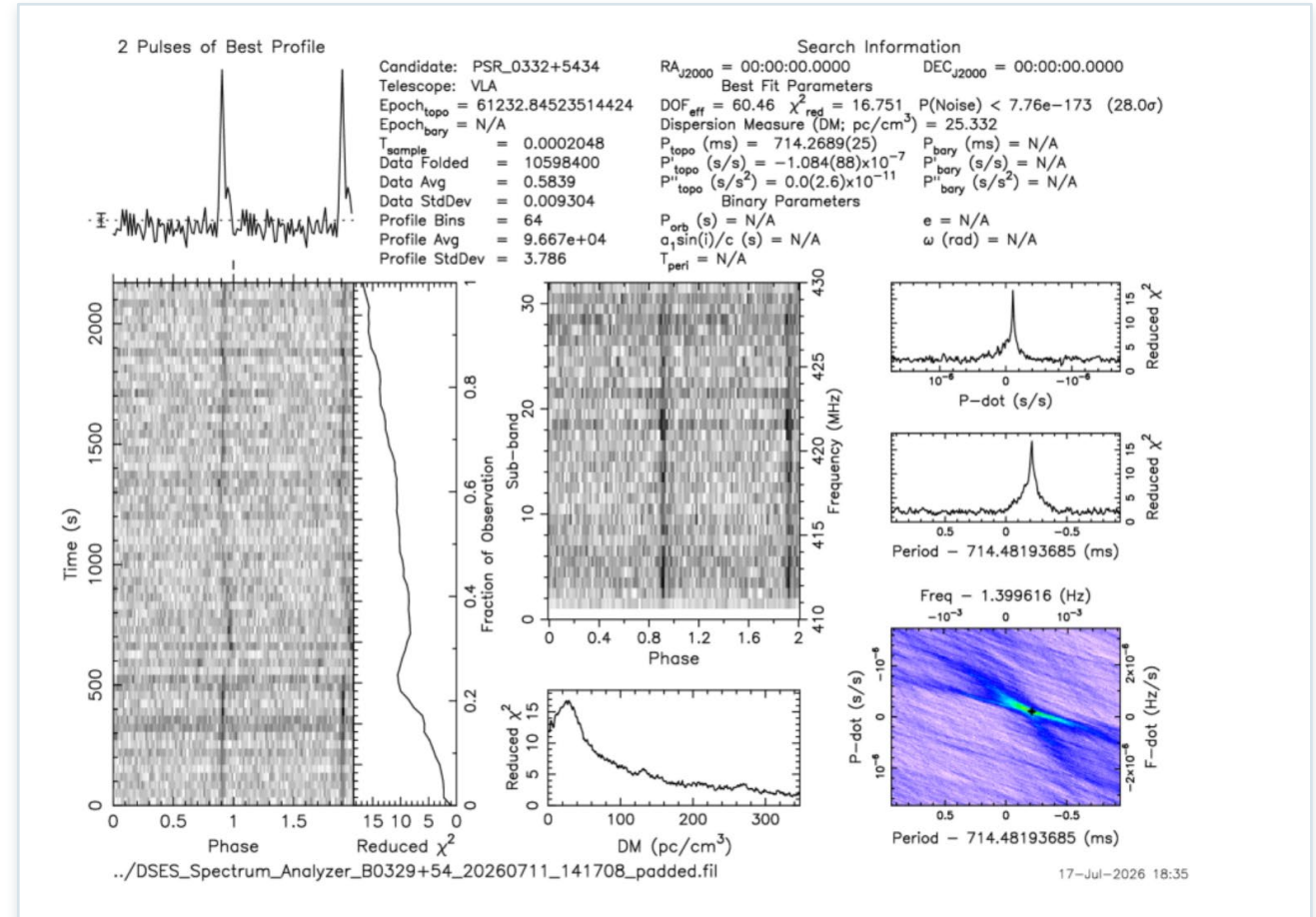
Targets were chosen live — the software computed current altitude/azimuth for the catalog pulsars from the Haswell site. Most southern sources never rise there, which made the transiting B0950+08 the natural second pick.

B0329+54 — a textbook detection

28.0 σ

probability the signal is noise: $< 8 \times 10^{-173}$

- Textbook single-peaked profile at the catalog period of 714.5 ms, coherent across the full 36.6 minutes
- Detection peaks at dispersion measure ≈ 25 — catalog value 26.8 — and falls off toward zero
- In the field it stood at $\sim 20 \sigma$; RFI cleanup and one timebase repair recovered the rest — that story is the next slide



Gap-bridged, RFI-cleaned ephemeris fold — the full 36.6-minute integration (2026-07-17 re-analysis)

One review comment was worth 5.6σ

1

The comment

Dan Layne (ADOCY), reviewing the trip report: the fold shows a phase drift it shouldn't have

2

The forensics

A rigid no-search ephemeris fold: smooth 1.2-rotation drift plus one +0.22-rotation step — parts in 10^4 , five orders beyond any physics. Our clock, not the sky

3

The culprit

USB sample drops during recording silently shorten the .fil's sample-count clock — the app showed overflows live but didn't count or log them

4

The repair

0.558 s of noise padded at the step: $22.4 \sigma \rightarrow 28.0 \sigma$, single textbook profile, DM back at 25.3 — diagnosis confirmed by repair

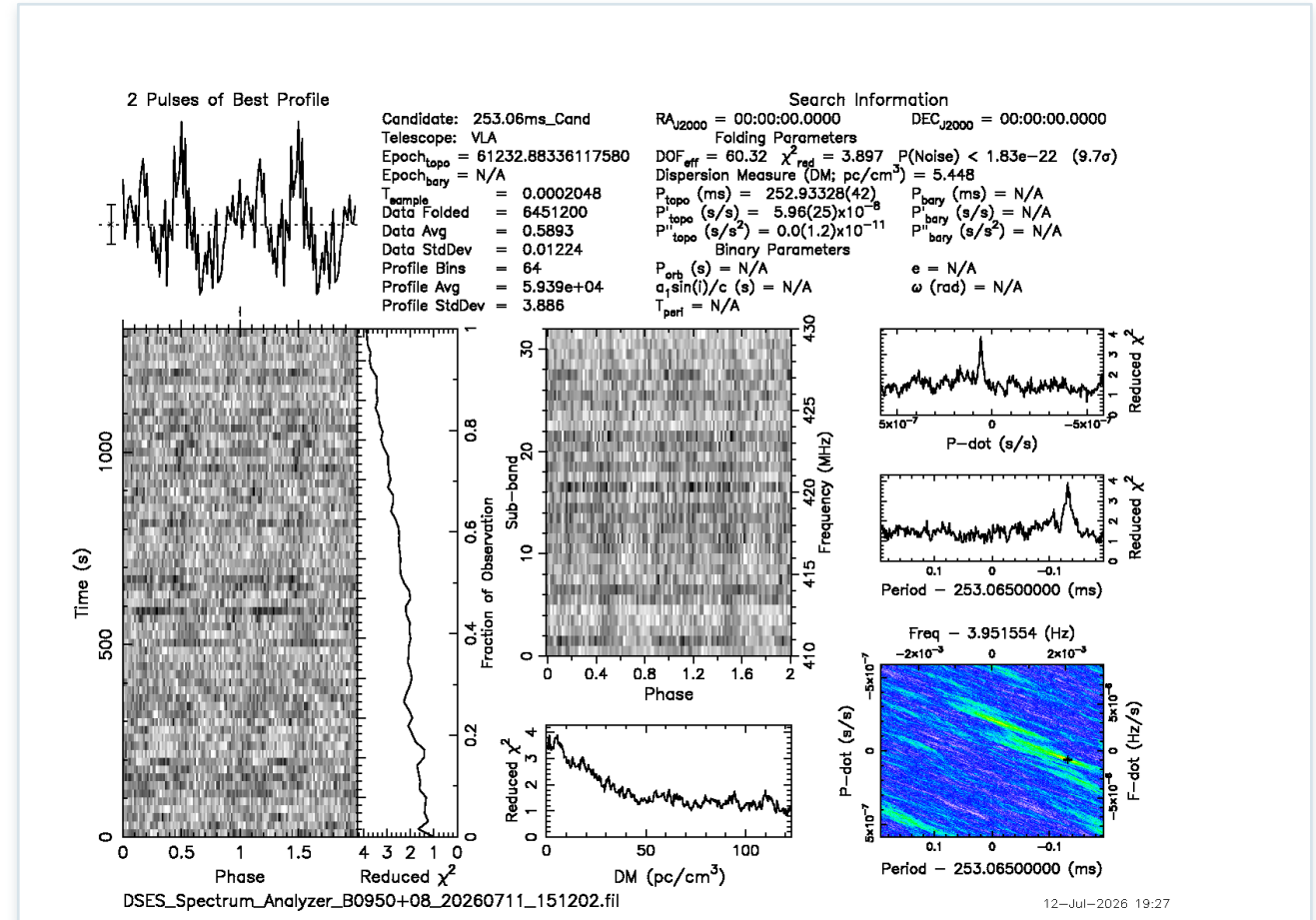
Return on criticism. Overflow logging and automatic gap-padding are now queued on the roadmap — future recordings keep an honest clock.

B0950+08 — a strong candidate, honestly framed

$\sim 9 \sigma$

after cleanup, the best fit lands on the catalog values

- Clear peak at the pulsar's 253 ms period when folded at the known ephemeris ($\sim 9.7 \sigma$ as recorded)
- With RFI masked and the timebase repaired, the search converges on $P = 253.05$ ms and $DM \approx 2.9$ — the catalog values — exactly how a real detection behaves
- But it is noise-limited: at 20 MHz bandwidth DM 2.97 is indistinguishable from zero, and $\sim 9 \sigma$ leaves no room for decisive sub-tests — a strong candidate, not yet a publishable detection
- The remaining gains are observational: a 90-minute track ($\approx 20 \sigma$), several sessions to catch scintillation maxima, wider bandwidth



Fold at the known 253 ms period — 22.2-minute observation

Real — and independently reproducible

Dispersion: the celestial fingerprint

- Interstellar plasma delays lower frequencies — a sweep imposed by light-years of travel
- Detection strength peaks at the catalog dispersion measure and falls toward zero
- Terrestrial interference shows no dispersion at all
- **One curve separates a genuine pulsar from RFI**

Two platforms, two toolchains, one answer

- macOS — PRESTO v5.0.2, re-run on v6.0.0
- Windows — PRESTO v6.0.0 under WSL (a new DSES capability)
- Identical fold commands, independently built stacks

B0329+54 probability-of-noise

4.49×10^{-90} Windows **4.5×10^{-90}** macOS

Every fold now produces a standardized PDF diagnostic stored beside the recording — validation minutes after capture, on whichever machine is at hand.

The case of the headless Raspberry Pi

1

Symptoms

Dropdown menus dead,
Help menu blank, window
opens tiny in a corner —
on the field box, mid-
session

2

Diagnosis

Remote SSH into the site
over Tailscale — while the
trip continued

3

Root cause

No monitor attached: the
graphics system reports a
0×0-pixel screen,
collapsing every popup
and confusing window
placement

4

Fix — in v1.1.6

Launcher synthesizes a
virtual monitor when none
is connected; the app
refuses to reposition
windows against an empty
screen

Not an application bug. A display-configuration condition that had silently affected every prior software version — found because real hardware was running at a real site.

Field wish-list → v1.1.6 in two days

Spectrum Analyzer v1.1.6 — cut and published July 12, one day after the observing session:

Sweep mode

Stepped wide-spectrum scan for RFI surveys beyond the radio's instantaneous bandwidth — retunes across a range (e.g. 100–1000 MHz) and stitches one wide trace, with cross-pass max-hold.

Recording, finished

Source name stamped into the filename and the .fil header — with RA/Dec derived from the pulsar designation. Red REC indicator, elapsed time, timed auto-stop. July 11's files had to be renamed and patched by hand; never again.

Headless fixes

The virtual-monitor launcher and empty-screen safeguards from the previous slide, plus updated documentation and operating guide.

Epilogue: one day after release, the field box caught a packaging defect desktop testing had missed — Windows line endings in the Unix launcher script. Rebuilt and republished within hours, with two independent build-pipeline safeguards added. Real installations in the field earn their keep.

v1.1.7 six days later — the display was hiding signals

1

The report

Ray (AAOL): on the same hardware, other SDR software shows weak signals ours doesn't

2

The root cause

The display FFT examined only the newest FFT-length of samples each screen tick — about 0.3% of the stream at 20 MS/s. The other 99.7% never reached the screen

3

The fix

Welch-average every sample between screen updates; true per-block peak/min hold detectors; an adaptive CPU budget so slower machines shed coverage gracefully

4

The result

A razor-flat noise floor with sub-dB scatter — weak signals stand clear. Verified live on the B210; released July 18

Recordings were never affected — the filterbank path always processed every sample; this was display-only. Two field reports, two releases, seven days.

A documented upgrade queue

Pulsar visibility planner

“what’s up now?” — pulsars above the horizon with az/el, flux, and time left; a pick fills the recording Source field

Quick-look during recording

a draft fold on a snapshot of the data so far — without interrupting a multi-hour capture

Self-contained fold PDFs

commands, results table, and plain-language verdict inside every chart PDF — nothing to chase down later

System-1 steering integration

pick a pulsar in the app and the dish knows where to point — awaiting the steering team’s API

One-click post-processing

“is it good?” — automatic RFI mask + catalog fold when a recording ends; sigma/DM results card and a plain verdict

Timebase integrity

count and log overflows; automatic gap-padding keeps the .fil clock honest — the peer-review finding, fixed at the source

Audio demodulators for RFI ID

click a signal and listen — AM/NFM/WFM/SSB/CW; an AI mode-classifier to follow

ezRA drift-scan format

record ezCol-compatible files for the open-source hydrogen-line drift-scan suite

8

features specified and committed to the Society’s shared roadmap

$\approx 20 \sigma$

expected from a 90-minute B0950+08 re-observation once the timebase fix ships

Plus a site-computer upgrade to multi-core x86-64 — enabling PRESTO post-processing right at the dish.

Training the radio-astronomy interest group starts on this baseline — a roster of members is signed up.

Three weeks later: the road-ahead list became a release

v1.1.8 — released August 3

- **Observation presets:** one pick — “Pulsar: L-band” — sets every science-critical parameter to the validated geometry; a live readout shows time resolution, DM smearing, disk rate.
- **Recording integrity:** dropped-sample gaps measured from radio timestamps and zero-padded exactly, live — the 28σ gap-repair lesson, now automatic (verified: 0 samples lost at 16 MS/s).
- **Automatic post-processing:** recording stops → PRESTO runs → a self-contained PDF: fold chart, commands, numbers, and a plain-language verdict (incl. “that’s terrestrial, not a pulsar”).
- **ezRA drift-scan format:** the same app writes ezCol-compatible files for Ted Cline’s ezRA suite — the DSES 60-ft dish is doing hydrogen-line drift scans with it today.
- **True hardware sample rates:** B210 to its real 61.44 MS/s ceiling, master-clock-clean presets.

v1.2.0 — landed this morning

- **Full UI redesign:** menus, six dockable panels, one-click observation setup.

WHAT WE PROVED AT HASWELL

- 1 **Non-experts detected B0329+54 at 28σ — plus a strong second candidate — with software the Society wrote, controls, and can teach.**
- 2 **The whole data path — antenna → SDR → filterbank → PRESTO — is validated end to end, reproduced across platforms, and hardened by peer review.**
- 3 **The barrier is now a slope: two releases shipped from field feedback in a week, and the upgrade queue is written down.**

Standing on the work of Dr. Richard Russel (AC0UB) and Dan Layne (AD0CY) — whose review of the report made the headline number stronger.

Field team: Ray Uberecken (AA0L), Anne Haney (WOZDW), Richard Hambly (K0GD).

Development, field diagnosis, and data processing accelerated by AI-assisted engineering (Claude Code).

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